

BAURods

Global Sensor Characterization

Technical report on the integrated 3 × 3 visuotactile sensing skin

Prepared 6 October 2026 | Manual acquisition: 8 August 2026

BAURods demonstrates useful **model-assisted contact discrimination and nine-region localization** across the integrated 3 × 3 sensing skin. The strongest supported system-level results are **98.50% contact balanced accuracy**, **84.40% localization accuracy**, and **0.499 N force MAE**, from retrospective nested grouped evaluation. These results complement the global physical response analysis of **5,357 retained repeated presses across 72 recordings**. The physical analysis establishes measurable optical response and known-contact spatial discrimination; the learned models quantify what can be inferred from richer optical features within their evaluated interval.

The model results use original, unannotated video features and six complete held-out TEST groups, with equal total weight for each recording. Force and localization use 7,154 eligible frames from 54 primary press recordings within 0.05-3 N. Contact adds 1,352 no-contact frames from 11 recordings (8,506 frames; 65 recordings total). These are correlated frame observations, not 8,506 independent contact events. The model protocol excludes 18 replay-designated recordings included in the broader 72-recording pulse analysis. Both analyses trace to the same manual archive; they are complementary evaluations, not independent replication. The recordings were collected on 8 August 2026; 9 September refers to saved model experiments, not new acquisition.

The learned pipelines were evaluated offline and were not installed into the live acquisition GUI. The archived 5-V acquisition gate remains a separate diagnostic with 0.597% press recall. Its result does not describe the later learned detector. Conversely, the learned detector's frame scores cannot be substituted for that gate's event-level performance or presented as prospective sensor validation.

Model-assisted sensing performance

BAURods model-assisted performance on held-out recordings

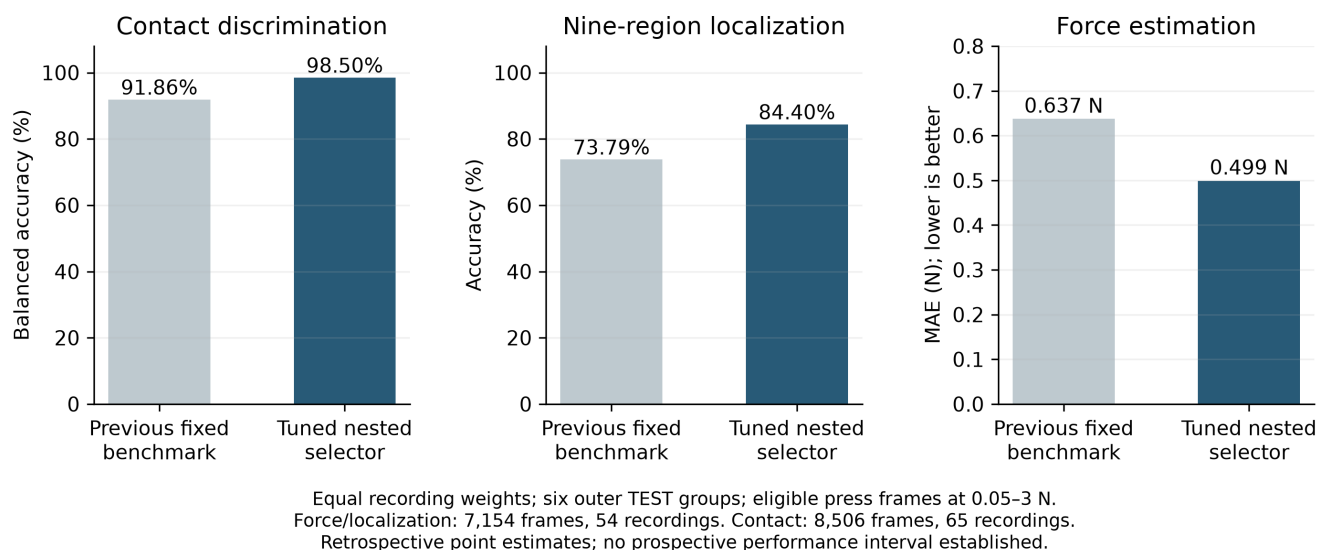


Figure 10. Model-assisted performance on matched held-out frames. Six outer TEST groups; equal recording weights; eligible press force 0.05–3 N. Fixed-family comparators were identified retrospectively; the tuned result evaluates the nested selector.

Table 1

Model-assisted characteristic	Previous fixed model	Tuned nested selector
Contact balanced accuracy	91.86%	98.50%
Contact accuracy	91.91%	97.51%
Contact recall	91.93%	97.00%
Contact specificity	91.80%	100.00%
Contact macro F1	0.871	0.958
Nine-region localization accuracy	73.79%	84.40%
Localization macro F1	0.740	0.848
Joint correct contact and region	70.50%	83.61%
Force MAE	0.637 N	0.499 N
Force RMSE	0.770 N	0.632 N
Force R ²	-0.030	0.306

The fixed comparators are Extra Trees for force, logistic regression for localization and Random Forest for contact, selected retrospectively from the fixed benchmark. The tuned headline is the complete nested selector, not the most favorable family chosen after inspecting outer scores. Force MAE decreased by 21.68%, localization accuracy increased by 10.62 percentage points, and contact balanced accuracy increased by 6.64 points on matched held-out frame identities. These gains are retrospective and were not uniform across force bins or locations. Model report, Tables 9–20 [S1].

Global Characterization of the BAURods Visuotactile Sensing Skin

Table 2

Sensor characteristic	Global baurods result	Interpretation / scope
Valid segmented physical presses	5,357 events, clustered in 72 press recordings	Force-only segmentation; not all statistically independent
Sensing area	All nine locations of one 3 × 3 BAURods skin	Taxel is a secondary grouping variable
Acquisition scope	83 recordings; one session S3 on 8 August 2026	Four baseline/settings blocks, not four independent days
Tested peak-force range	1.1512-23.3004 N	Retained presses; not maximum sensor capacity
All raw reference-force range	-0.2867-23.3004 N	Negative unloaded offsets included
Optical-response range	0.7071-8.5698 V digital units	Peak maximum ROI mean positive delta-V from magnified images
Global force-response relationship	$S = 0.90855 + 0.12106 F$	Descriptive pooled association; no intrinsic calibration established
Global slope / apparent sensitivity	0.12106 V digital units/N; 95% CI 0.08885 to 0.14901	Processing- and protocol-dependent; not intrinsic material sensitivity
Model fit	$R^2 = 0.2229$; RMSE = 0.7045 V	In-sample, press-level
Validation	Recording-held-out $R^2 = 0.1934$; RMSE = 0.7178 V	Model selection used this CV; no final untouched test set
Repeatability proxy	Residual SD = 0.7046 V; IQR = 0.8394 V	Conditional dispersion under manual loading; not controlled metrological repeatability
Baseline variability	Global mean = 0.8980 V; RMS temporal SD = 0.0437 V	11 recording-equal no-contact controls
Statistical force detection threshold	Not quantitatively assessed	No validated replicated low-force staircase; segmentation excludes excursions <0.5 N
Lowest observed detected press peak	1.8699 N	One observed peak, not a reliable threshold
Configured optical decision criterion	5 V digital units	Already present in acquisition software; not 5 N
SNR proxy	Median = 18.406; IQR = 23.600	(Peak response - global no-contact mean) / RMS temporal SD
Response time	Not quantitatively assessed	Sparse samples and narrow-band magnification prevent intrinsic latency inference
Recovery time	Not quantitatively assessed	Repeated presses generally retain residual load and have no stable release baseline
Hysteresis	Not quantitatively assessed	No controlled comparable quasi-static loading/unloading protocol

Sensor characteristic	Global baurods result	Interpretation / scope
Operating limit / saturation	Not quantitatively assessed	Tested force maximum is not a failure or saturation limit
Archived-gate contact accuracy	0.801% (43/5368)	Unequal positive/negative units; report recall and specificity alongside
Archived-gate contact recall	0.597% (32/5357)	Existing 5-V rule
Archived-gate specificity	100.00% (11/11)	11 one-second no-contact controls
Archived-gate localization accuracy	0.597%	All valid press events; below-threshold outputs count as failures
Archived-gate localization macro F1	0.01135	Nine target classes; no-detection column retained
Known-contact location discrimination	94.307% accuracy; macro F1 = 0.94437	Offline ungated argmax diagnostic, not archived end-to-end performance

1. Global Response Definition

The acquisition pipeline uses OpenCV HSV **V (brightness)**, not hue or saturation as the response. For ROI i at time t , it forms $I_i(t) = \text{mean}_p[\max(V_{\text{magnified}}(p,t) - B_i(p), 0)]$, where B is the captured per-pixel baseline median after the software's integer conversion. The frame-level global metric is $G(t) = \max_i I_i(t)$, exported as `dominant_intensity`. For press e , the primary response is $S_e = \max_t(t \text{ in } e) G(t)$. The reference is $F_e = \max_t(t \text{ in } e) F_{\text{raw}}(t)$, using the higher-rate load-cell samples. Units are **V digital units**, based on 8-bit image values, not volts, lux, physical light power or generic HSV units. Force is in newtons.

This choice follows the existing algorithm: it selects the ROI with the greatest mean positive delta-V and emits its location when the value reaches 5 V units. The metric is independent of the known pressed location, does not weight unequal ROI areas by pixel count, and gives one sensor-level response for a press anywhere on the array. The positive pixel subtraction already corrects against the captured baseline. A positive unloaded noise floor remains because negative differences are clipped and a maximum is selected. We retain this floor and an intercept rather than forcing zero response at zero force.

All 83 archived CSV exports were computed from color-magnified images, with amplification 30 and a 1.1-1.2 Hz temporal passband in the saved configuration. Therefore these are properties of BAURods plus this acquisition/processing configuration. They are not raw-light material constants. Original-video feature-store values are used only as a separately labelled diagnostic after verifying the source ZIP identity, every input Parquet hash, frame count and force alignment. They do not replace the archived software results.

Alternatives were compared without selecting the best-looking correlation: maximum across ROIs (primary), sum of ROI means (includes distributed noise and common-mode changes), response in the known pressed ROI (requires ground truth), response nearest the force peak (timing diagnostic), and original-video response (source diagnostic). `response_definition_comparison.csv` preserves these comparisons. The sum here is a sum of ROI **means**, not a sum of all pixels. Force and optical event peaks can occur at different times; their association describes pulse amplitudes, not a synchronized pointwise constitutive law. The nearest-force-peak response has R^2 0.0274, substantially below the paired pulse-amplitude model; timing is a material qualification. The original-video diagnostic has R^2 0.2317. Pipeline definitions can be inspected in feature extraction and localization.

Experimental unit and press segmentation

The operator confirmed that the rapid loading pulses were separate finger presses. The raw load-cell trace identifies candidate peaks with prominence at least **0.5 N** and a minimum sample separation corresponding to **0.15 s** (two raw samples at the observed cadence). This allows fast pulses to remain distinct rather than forcing nearby presses into one interval. Adjacent force minima define non-overlapping optical windows [start, end). A retained event must rise and fall by at least 0.5 N, have a peak above the recorded 0.05-N force contact criterion, contain at least three valid camera samples, and have a camera sample within 0.15 s of the force peak. The selected force peak must also be the maximum within its interval; unresolved intervals are rejected. **2000 candidates were excluded** with reason codes. Optical magnitude and localization correctness never determine eligibility. The camera-sample requirement preferentially removes the fastest pulses and is an explicit sampling limitation.

The 0.5-N segmentation prominence is an analyst-defined pulse-separation rule, not a sensor detection threshold. The comparison at 0.25, 0.5 and 1.0 N gives 5,356-5,443 events, with pooled slopes 0.11973-0.12246 V units/N. This limited sensitivity check supports the stability of the main descriptive conclusion to prominence choice; it does not certify the count. A four-page segmentation atlas is supplied for all 72 press recordings. Peaks with insufficient unloading, unresolved multiple pulses, very small force excursions and recording-edge partial presses cannot be certified individually. No claim is made that the inferred count equals a manually annotated ground-truth press count.

The archive contains **29,318 camera rows** and **42,568 raw load-cell samples** in 83 files. There are 72 press recordings (eight per location) and 11 no-contact controls. Eight NOCONTACT files retained a Press interaction label; their filenames and measured $|F| < 0.05$ N identify them as controls. The remaining three are explicitly labelled none. All frame validity checks pass. There are four captured baseline/settings blocks but only **one session label S3**, one sensing skin and one collection day. Recordings, not frames or alleged separate acquisition days, are the resampling clusters.

The pooled fit gives every retained physical press equal weight. Confidence intervals resample complete recordings **within each taxel**, preserving the nine fixed sensing locations (2,000 bootstrap replicates, seed 20261005). Taxel is not treated as nine separate sensors or as nine independently manufactured specimens. The intervals describe recording-to-recording uncertainty within this experiment; they do not establish between-device or between-day reproducibility. They condition on recorded force and do not include independently assessed load-cell calibration uncertainty. Model validation holds out whole recordings, never random frames or presses from the same recording.

2. Tested Force and Response Range

Retained press peaks span **1.1512-23.3004 N** and their global responses span **0.7071-8.5698 V units**. The full raw reference-force series spans -0.2867-23.3004 N, including slightly negative unloaded offsets. Negative offsets were retained as source values; no speculative per-press zero adjustment was made. Mean peak force was 7.8125 N (SD 3.1172); mean optical response was 1.8543 V (SD 0.7993).

These are **tested ranges**, not maximum sensing limits. Different force coverage, duration, loading rate and baseline/settings blocks prevent attribution of pooled flattening or compression to material saturation. A pixel-level saturation_warning is a separate image-clipping diagnostic; it is not proof of force saturation. 0 retained presses included an exported pixel-saturation warning.

Table 3, part 1

Measure	Unit	Count	Mean	SD
Peak force per press	N	5357	7.8125	3.1172
Peak global optical response per press	V digital units	5357	1.8543	0.79928
Force excursion per press	N	5357	5.2902	2.7528
Force-defined pulse duration	s	5357	0.57402	0.26208
Peak excess / pooled baseline temporal SD	dimensionless	5357	21.875	18.282
No-contact recording mean global response	V digital units	11	0.89797	0.033158
Captured baseline global raw V mean	V digital units	4	28.238	2.2241

Table 3, part 2

Measure	Median	Q1	Q3	Iqr
Peak force per press	7.3306	5.4706	9.7146	4.2441
Peak global optical response per press	1.7026	1.2434	2.2751	1.0317
Force excursion per press	4.96	3.0963	6.9856	3.8893
Force-defined pulse duration	0.54896	0.44851	0.64321	0.1947
Peak excess / pooled baseline temporal SD	18.406	7.9006	31.5	23.6
No-contact recording mean global response	0.90729	0.87513	0.91414	0.039006
Captured baseline global raw V mean	27.43	26.882	28.787	1.905

Table 3, part 3

Measure	Minimum	Maximum	CV (%)	Mean CI lower
Peak force per press	1.1512	23.3	39.901	7.2146
Peak global optical response per press	0.70712	8.5698	43.103	1.696
Force excursion per press	0.54086	19.115	52.035	Not available
Force-defined pulse duration	0.26113	3.4906	45.657	Not available
Peak excess / pooled baseline temporal SD	-4.3654	175.48	83.575	Not available
No-contact recording mean global response	0.85497	0.96365	3.6925	0.87944
Captured baseline global raw V mean	26.615	31.475	7.8764	Not available

Table 3, part 4

Measure	Mean CI upper
Peak force per press	8.316
Peak global optical response per press	2.0159
Force excursion per press	Not available
Force-defined pulse duration	Not available
Peak excess / pooled baseline temporal SD	Not available
No-contact recording mean global response	0.9169
Captured baseline global raw V mean	Not available

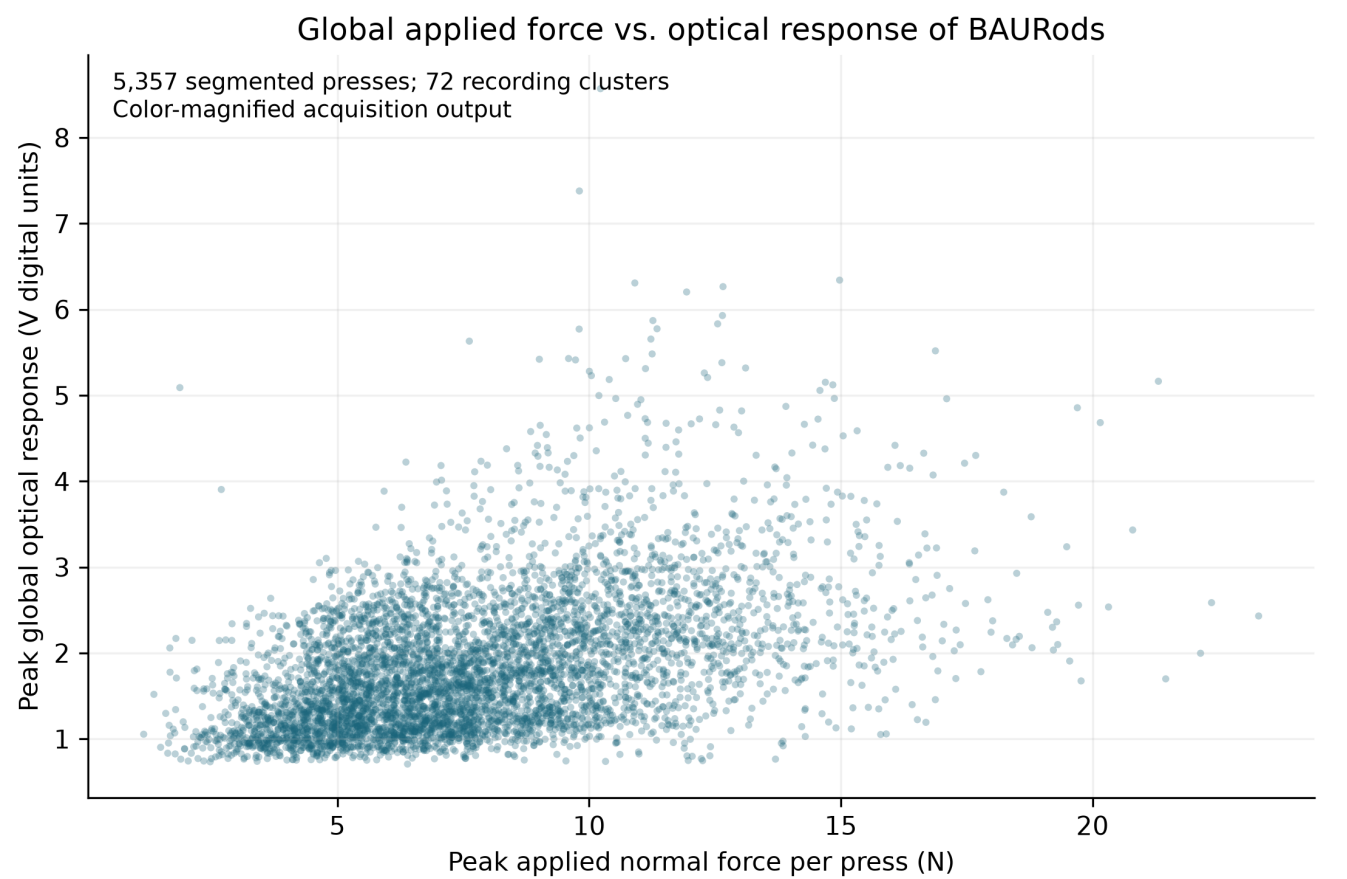


Figure 1. Global applied peak force and optical response of the complete BAURods skin. Each point represents one retained force-segmented press; observations remain clustered in recordings.

3. Global Force-Response Behavior

The selected parsimonious descriptive model was **linear**, with equation $S = 0.90855 + 0.12106 F$, where F is press peak force in N and S is peak global response in V digital units. In-sample R^2 was **0.2229** and RMSE was **0.7045 V**. Recording-held-out R^2 was **0.1934** and RMSE was **0.7178 V**. Intercept CI was 0.73517-1.09366. Parameters and intervals are exported separately.

Linear, quadratic, logarithmic and power-law models were compared with a constant-response benchmark. Fits minimize squared errors in original response units; the power model is not selected by a log-transformed R^2 . The one-standard-error rule compares recording-equal cross-validation RMSE and favors the simplest adequate force-dependent form. The constant benchmark remains an adequacy check, even when a force-dependent line is presented for interpretation. No high-order polynomial or post-hoc force range was searched to increase R^2 . Validation here supports exploratory model comparison, not an untouched confirmatory performance estimate.

Table 4, part 1

Model	Count	Parameters	R^2	RMSE (V units)
constant	5357	1	0	0.7992
linear	5357	2	0.22292	0.70451
quadratic	5357	3	0.22478	0.70367
logarithmic	5357	2	0.21152	0.70966
power	5357	2	0.22228	0.7048

Table 4, part 2

Model	MAE (V units)	Held-out RMSE (V units)	Held-out R^2	Mean recording RMSE
constant	0.61046	0.81066	-0.028893	0.68336
linear	0.52771	0.71776	0.19342	0.57024
quadratic	0.5265	0.72044	0.18739	0.57019
logarithmic	0.53139	0.72223	0.18335	0.57513
power	0.52702	0.7181	0.19265	0.56898

Table 4, part 3

Model	SE recording RMSE
constant	0.03558
linear	0.033165
quadratic	0.033519
logarithmic	0.033194
power	0.033443

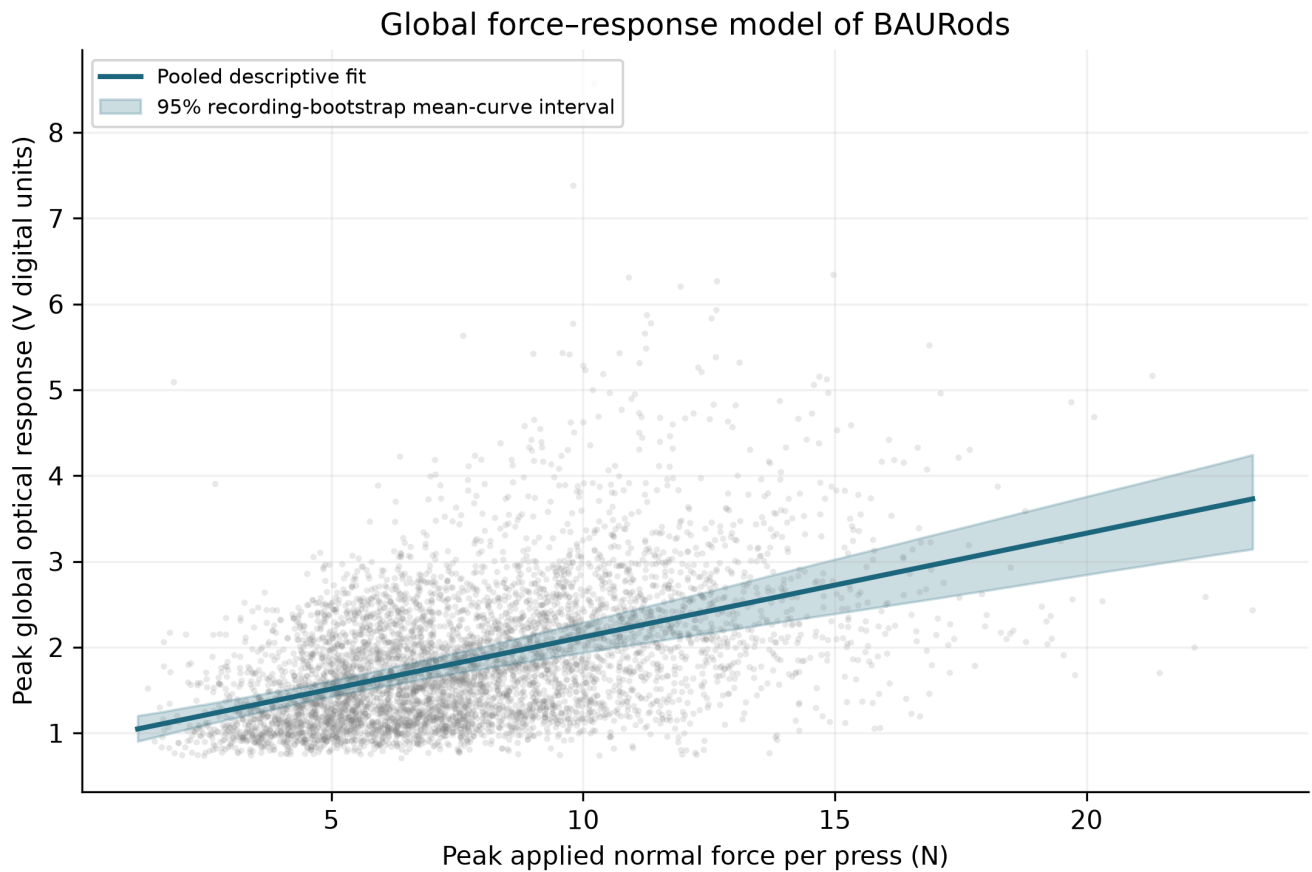


Figure 2. Global fitted force-response characteristic. The model describes archived optical response in V digital units; its slope is not an intrinsic material sensitivity.

4. Sensitivity and Linearity

Model-assisted force estimation

The learned optical-to-force pipeline attained **MAE 0.499 N, RMSE 0.632 N and R^2 0.306**, with signed bias +0.078 N, on recording-balanced held-out frames within 0.05-3 N. This provides a practical force-estimation result alongside the scalar response curve. The two regressions have different targets: the physical curve predicts V digital units from peak force; the learned model predicts N from multivariate original-video features. Their RMSE and R^2 must not be ranked against one another. The learned model does not establish intrinsic sensitivity or linearity.

The pooled MAE reduction of 21.68% supports improved retrospective estimation, with a clear remaining limitation: force MAE worsened in each of the four bins below 2 N. Below 0.5 N, tuned MAE was 1.241 N. Gains therefore concern the pooled evaluated distribution, not uniformly precise light-contact sensing. Model report, Tables 9 and 17 [S1].

Apparent optical slope

The pooled apparent slope was **0.12106 V digital units/N** (95% recording-bootstrap CI 0.08885-0.14901). This is a slope for the complete optical acquisition pipeline under this manual protocol. It does not establish intrinsic mechanoluminescent sensitivity or a transferable force calibration.

R^2 is a measure of explained dispersion within these observations, not proof of sensor quality. The residual plot, force-decile residual means and normal-probability diagnostic show departures that an R^2 alone conceals. Between-recording shifts and uneven force coverage can generate a pooled slope without establishing

within-location force linearity. No validated linear operating interval, linearity-error specification or force inversion accuracy is established by this experiment. Baseline/settings and taxel-adjusted models are secondary checks, not nine independent sensitivity claims.

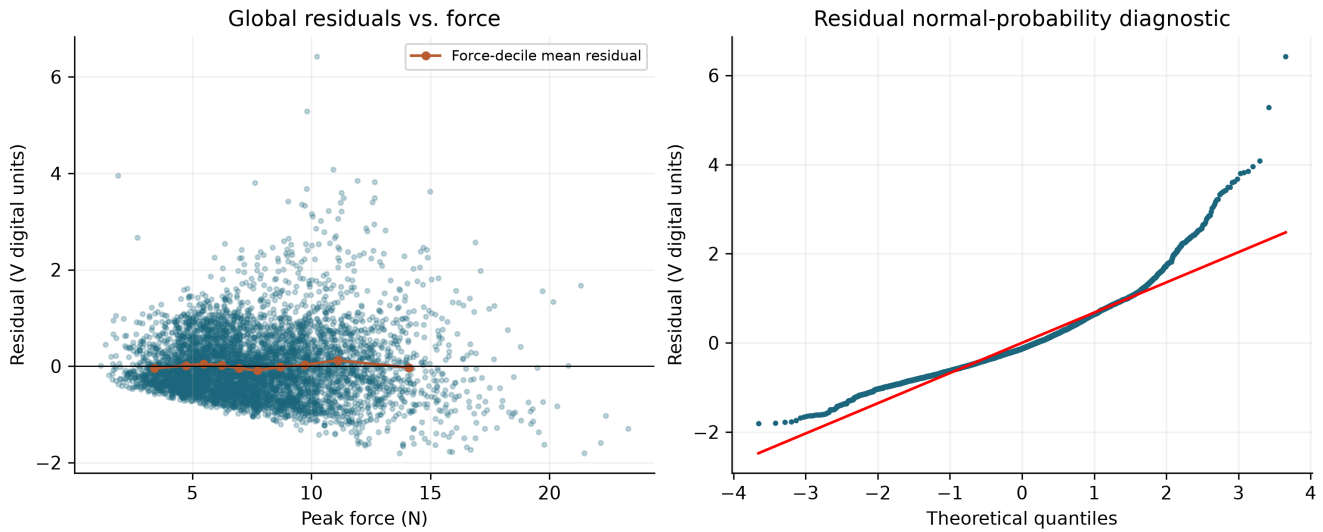


Figure 3. Residual diagnostics for the global force-response model.

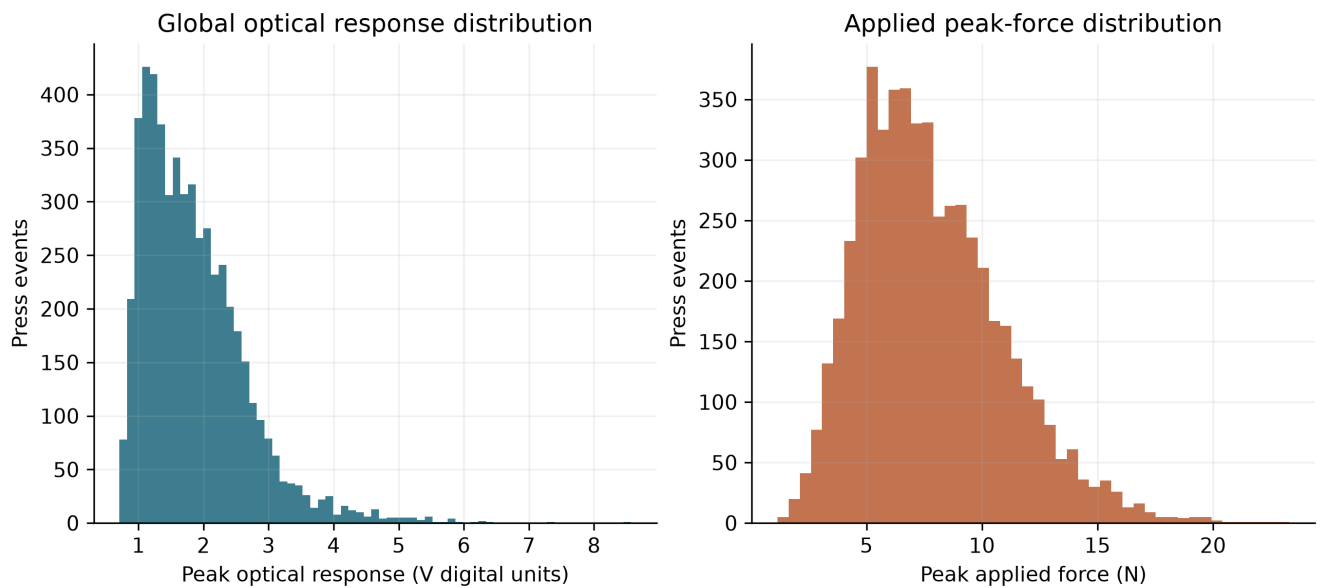


Figure 4. Distribution of global optical response across retained presses.

5. Repeatability

Residual SD around the global model was **0.7046 V**, with residual IQR **0.8394 V**. This measures force-conditioned dispersion under the manual protocol. It includes spatial differences, recording changes, loading rate, processing memory and model misspecification. It is not a controlled repeatability standard deviation at a fixed force and fixed location.

Empirical force tertiles supply comparable-force summaries without treating their raw response CV as a global repeatability claim. The pooled raw optical-response CV (43.10%) is descriptive only because force varies. Individual presses remain nested in recordings throughout uncertainty analysis.

Table 5, part 1

Force band (N)	Events	Recordings	Min force (N)	Max force (N)
(1.150, 6.121]	1786	69	1.1512	6.1204
(6.121, 8.85]	1785	58	6.1208	8.8488
(8.85, 23.3]	1786	51	8.8499	23.3

Table 5, part 2

Force band (N)	Mean response (V units)	Response SD (V units)	Response CV (%)	Residual SD (V units)
(1.150, 6.121]	1.4801	0.52252	35.302	0.50353
(6.121, 8.85]	1.7567	0.64077	36.475	0.63694
(8.85, 23.3]	2.326	0.92803	39.898	0.90934

Table 5, part 3

Force band (N)	Residual RMSE (V units)	Residual IQR (V units)
(1.150, 6.121]	0.50341	0.67671
(6.121, 8.85]	0.63835	0.83501
(8.85, 23.3]	0.90997	1.0957

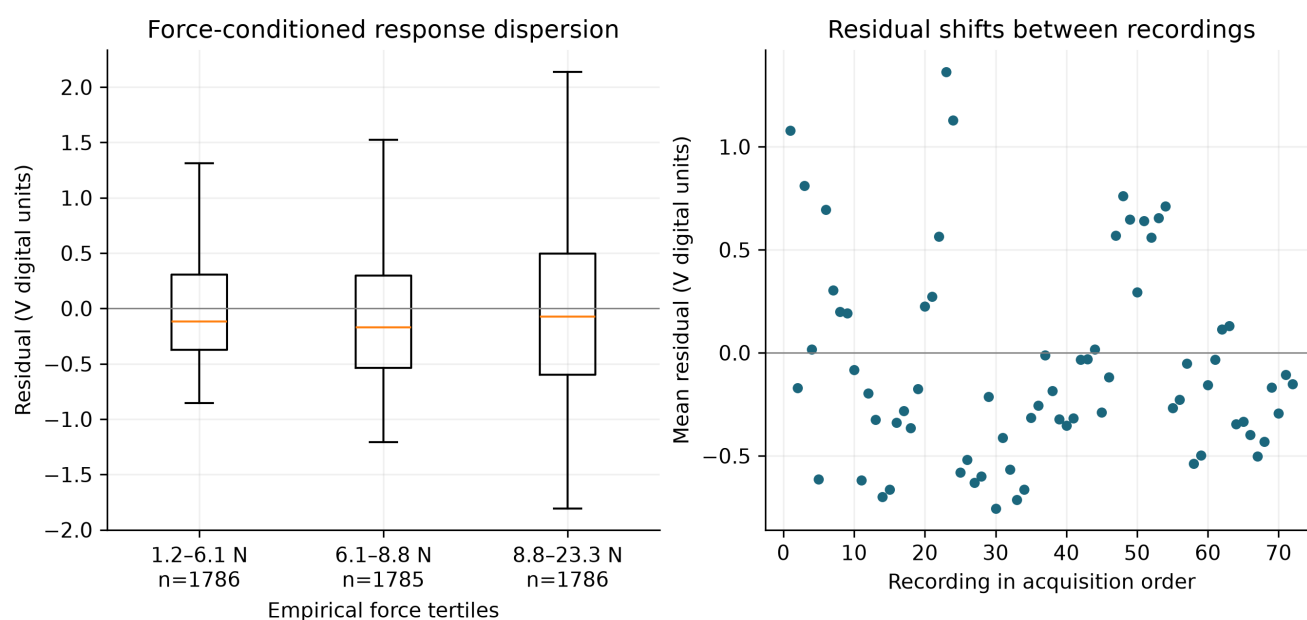


Figure 5. Force-conditioned response dispersion under manual pressing. This is a repeatability proxy, not controlled fixed-force metrological repeatability.

6. Baseline Stability

The 11 unloaded recordings produced a recording-equal global mean response of **0.8980 V** (95% recording-bootstrap interval 0.8794-0.9169). The root-mean-square within-recording temporal SD was **0.0437 V**, giving a temporal-noise CV of 4.87%. The SD of the 11 recording means was 0.0332 V. Temporal SD, between-recording SD and spatial dispersion of baseline image pixels are different quantities and are not substituted for one another.

The baseline table reports each recording's descriptive drift slope in V/min. It does not treat individual frames as replicates for confidence limits. Captured baseline image means are summarized once per unique baseline ID (four observations), not duplicated for every frame. Camera saturation and sharpness controls changed across baseline blocks, while autofocus and automatic white balance were enabled in the inspected configurations. Baseline shifts therefore cannot be assigned solely to sensor drift. No long-term or between-day stability estimate is possible from this single collection day.

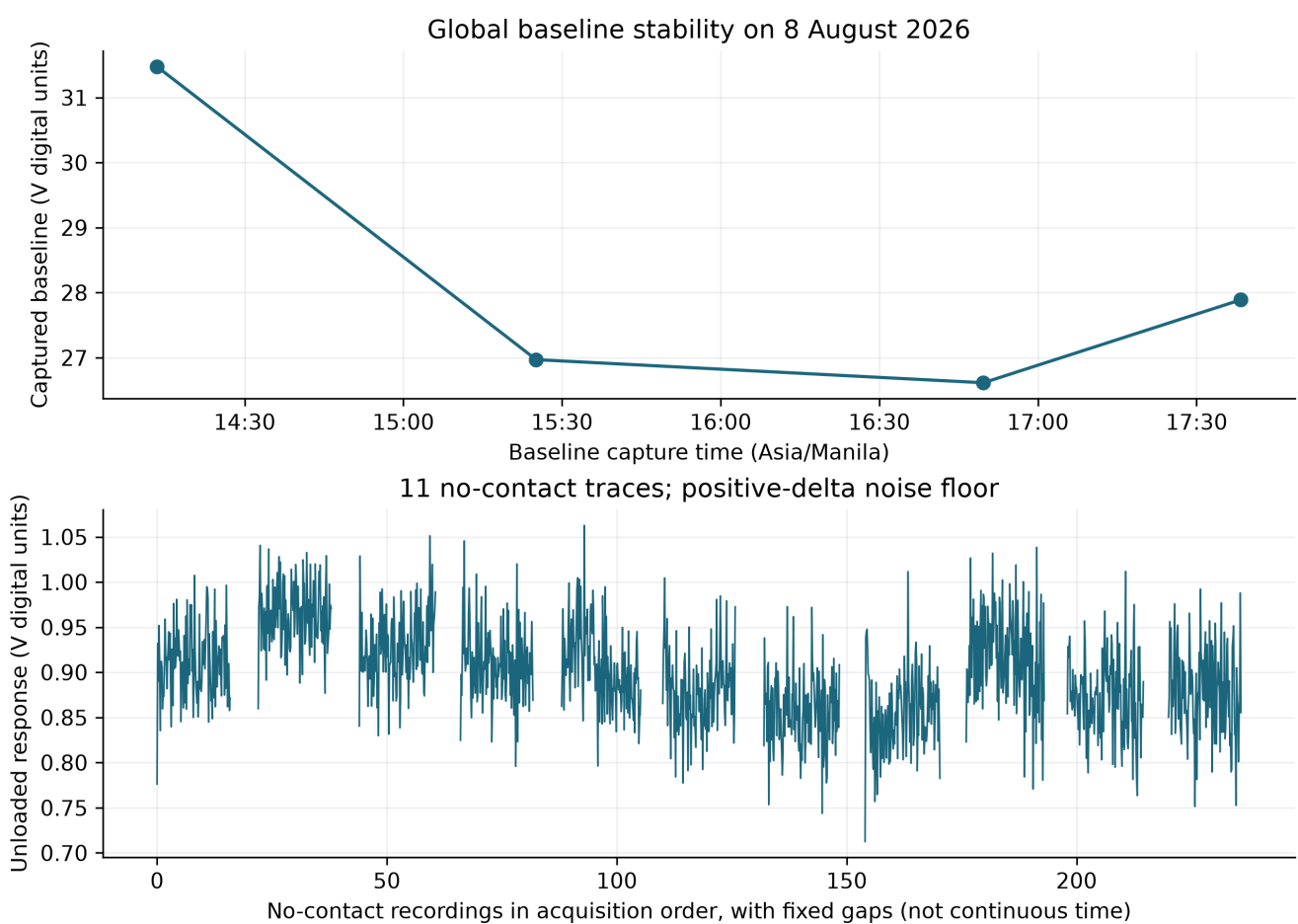


Figure 6. Recorded baseline stability and unloaded optical traces.

7. Detection Characteristics

Learned contact discrimination

The tuned nested detector achieved **98.50% balanced accuracy**, **97.00% recall**, **97.51% accuracy**, and **macro F1 0.9581**. Specificity and precision were both 100% on the observed dedicated controls. This is the principal model-assisted contact result: optical features distinguish eligible contact frames from unloaded recordings well under the archived conditions. The zero observed false-positive rate rests on only 11 no-contact recordings and does not establish zero deployment risk. No pooled tuned ROC-AUC is claimed because fold-selected score scales differ.

These metrics use recording-balanced frames within the model study's 0.05-3 N press interval. They do not count physical contact episodes or characterize onset/release. The final saved contact score threshold, 0.3268598318, belongs to its trained preprocessing, SVC score mapping and causal smoothing. It is neither an optical-brightness threshold nor a force limit of detection. Model report, Tables 11, 14 and 22 [S1].

Archived acquisition gate and baseline diagnostics

The existing acquisition rule reports a taxel only when the global response reaches **5 V digital units**. Applied once per retained press at its strongest global response, this rule detected **32/5357 presses (0.597% recall)**. Each unloaded recording contributes one centered 1-s control interval: **11/11** were negative, for specificity 100.00% (Wilson 95% interval 74.12-100.00%). Controls share session/baseline conditions, so this interval is conditional on treating recordings as replicates.

Precision was 100.00%, F1 was 0.01188, and accuracy was 0.801%. Accuracy is dominated by the large number of loaded events and is not a balanced estimate across contact/no-contact situations. ROC-AUC of the event response score was 0.93884; it is exploratory, uses only 11 control recordings, and compares pulse maxima with fixed-window maxima. The optical score is not a calibrated contact probability. Fixed 0.5-, 1- and 2-s negative-window checks are exported. No-contact frame counts are not used to inflate the negative sample size.

The lowest retained press peak that crossed the configured optical criterion was **1.8699 N**. That observation is **not** a force detection threshold. Reliability at low force cannot be established from opportunistic manual pulses, especially because event eligibility requires a 0.5-N excursion. The baseline mean + 3 temporal SD value, 1.0291 V, is an exploratory optical noise reference only. It was not used as a newly optimized classifier or inverted into a claimed force threshold. The maximum-across-ROI statistic, autocorrelation, temporal filtering and pulse-max selection preclude a simple Gaussian false-alarm interpretation.

The sensor-level SNR proxy is $(S_e - \text{mean_no_contact_G}) / \text{RMS_SD_no_contact_G}$. Its median was **18.406**, IQR 23.600. Negative values indicate a press peak below the pooled baseline mean. This linear ratio uses the same whole-sensor statistic for signal and noise; it is not a physical optical-power SNR and no dB conversion is asserted. No-contact controls cover only the later baseline blocks, so applying their pooled noise to earlier blocks is an exploratory approximation.

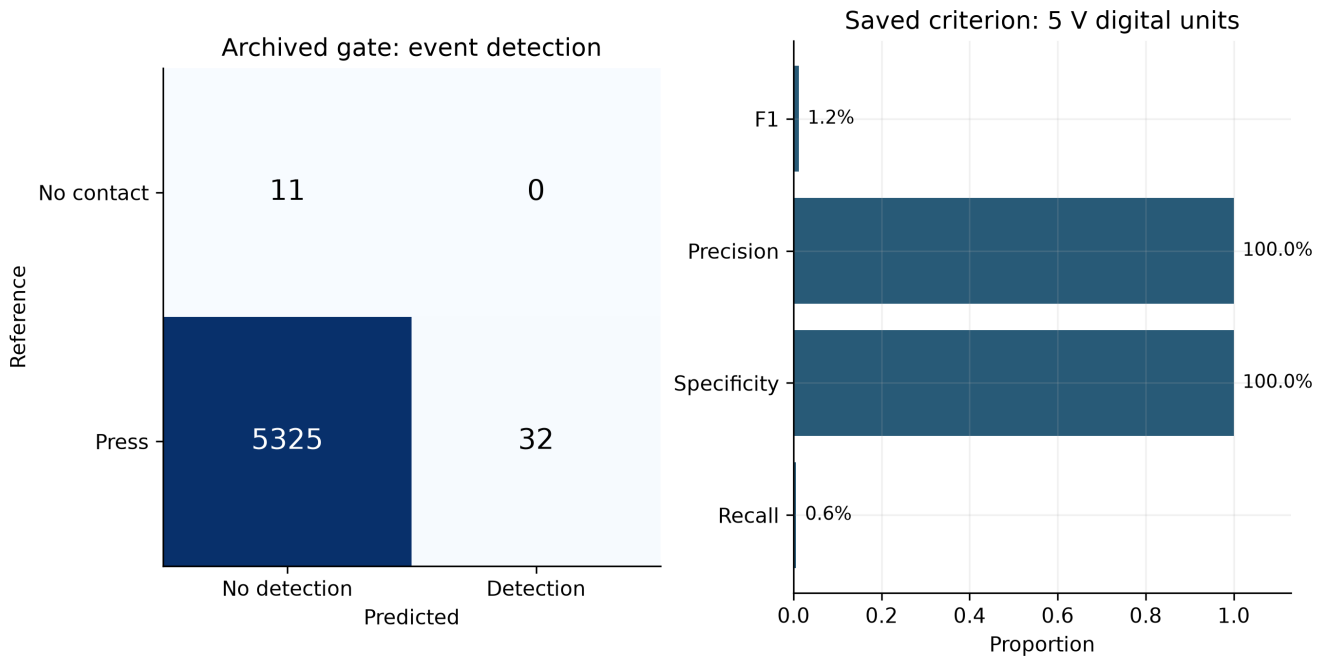


Figure 8. Archived 5-V acquisition gate: 5,357 retained presses and 11 no-contact recording windows. These event/window metrics are separate from the learned frame-level detector.

8. Temporal Characteristics

The model report supplies a separate **computational timing** result: offline stateful inference took a median **41.10 ms** and 95th percentile **44.46 ms**. That measurement includes optical-tabular preprocessing, causal state, the three main tasks and an alternate contact policy. It excludes camera acquisition, decoding, pixel extraction and display. Contact/localization output smoothing in the saved causal models uses a 1-s time constant; that parameter is not a measured onset or recovery time. The runtime measurement supports software feasibility only and does not fill the intrinsic response-time or recovery-time entries. Model report, Tables 21 and 23 [S1].

The median camera sample interval across recordings was **0.1301 s** (approximately 7.69 samples/s); the median raw force interval was 0.0919 s. The video container's nominal frame rate does not replace actual acquisition timestamps. Median force-defined pulse duration was 0.5490 s and the median number of camera samples per press was 4.

The recorded optical response passes through a narrow temporal magnification filter. Repeated manual loading often does not return to an unloaded plateau, and force and optical maxima may not align. Consequently intrinsic response time, 10-90% rise time and recovery time are **not quantitatively assessed**. `peak_time_difference_s` is preserved as a sampled force/optical peak offset, explicitly **not a response-time estimate**. Figure 7 illustrates actual force-defined pulses and optical sampling; it does not claim milliseconds of sensor latency. Unsynchronized photophysics or mechanical recovery cannot be separated from camera, filter and sampling effects with this protocol.

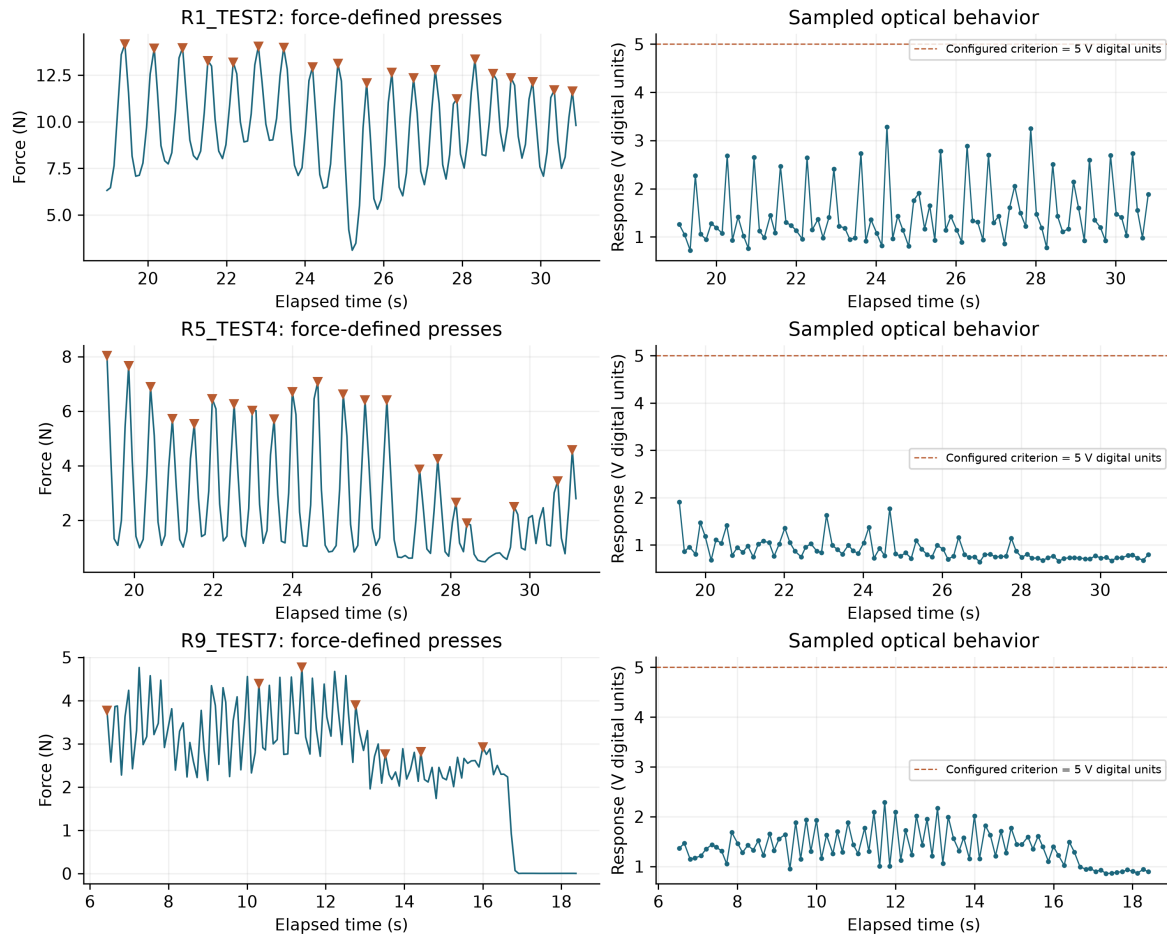


Figure 7. Sampled force and optical behavior during repeated manual presses. Temporal filtering and sparse samples prevent inference of intrinsic response or recovery time.

9. Hysteresis

Hysteresis could not be quantitatively characterized from the available experimental protocol. Force rises and falls during each finger press, but this alone does not establish comparable quasi-static loading and unloading trajectories. Uncontrolled rates, residual force, sparse optical samples and the 1.1-1.2 Hz optical filter confound any apparent loop with dynamic lag. A normalized loop area or maximum separation would therefore not identify intrinsic global sensor hysteresis. No automatic-calibration data were mixed into this manual-archive analysis.

10. Spatial Consistency

Taxel identity is retained only as a blocking/grouping factor. Recording-level mean global residuals differ across locations; these diagnose whether one pooled response remains representative. The range of location mean residuals is **-0.5814 to 0.4896 V**. The force-only linear model had R^2 0.2229; adding baseline block gave 0.3438; adding both baseline block and taxel gave 0.5336. These comparisons include recording-held-out RMSE in the accompanying table and should not be interpreted from in-sample R^2 alone.

Table 6, part 1

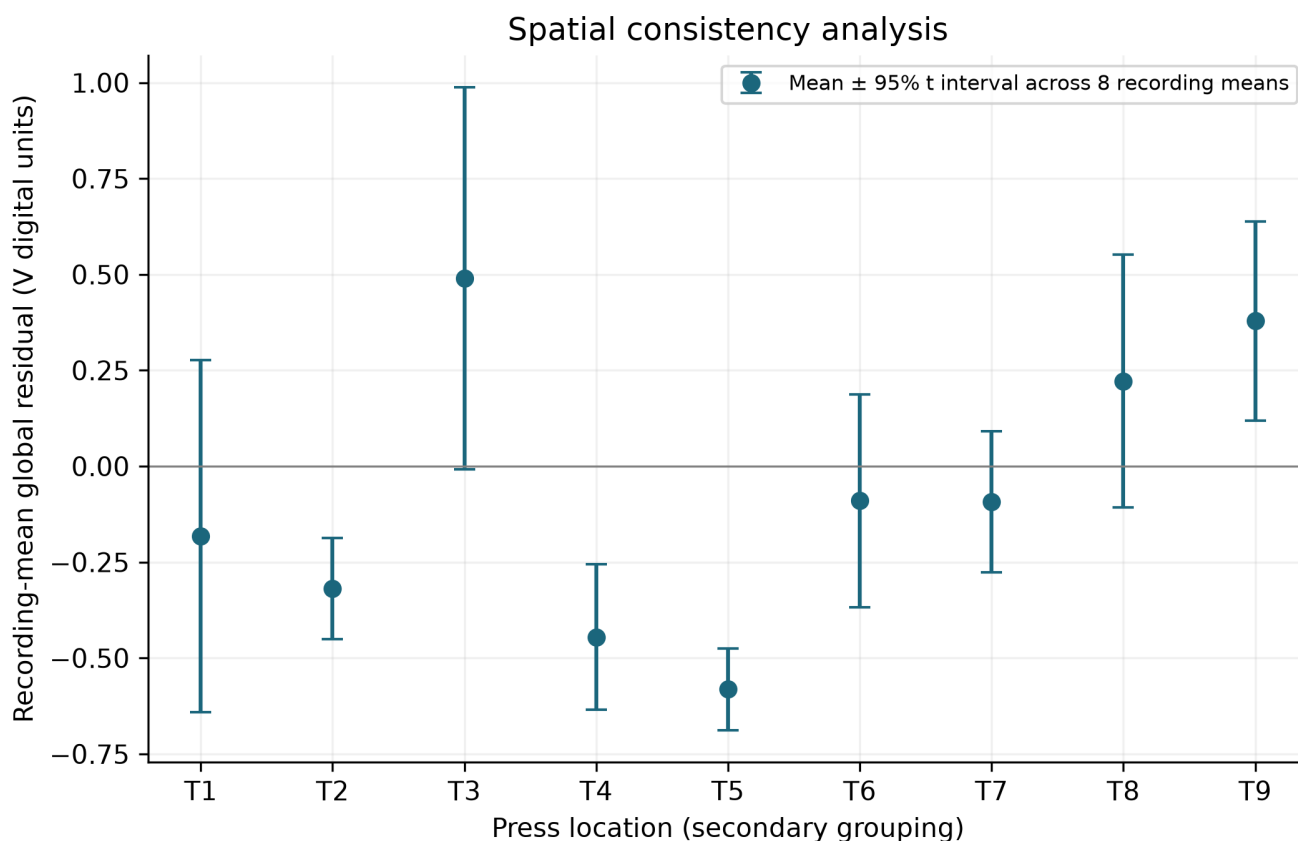
Model	Parameters	Force coefficient (V units/N)	R ²	RMSE (V units)
force only	2	0.12106	0.22292	0.70451
force and baseline block	5	0.098713	0.3438	0.6474
force and taxel	10	0.11897	0.44952	0.59296
force baseline and taxel	13	0.090627	0.53364	0.54578
within recording centered force	73	0.082331	0.67039	0.45883
equal recording weight linear	2	0.11616	0.21385	0.70861

Table 6, part 2

Model	MAE (V units)	Held-out RMSE (V units)
force only	0.52771	0.71776
force and baseline block	0.45639	0.67273
force and taxel	0.4165	0.66242
force baseline and taxel	0.38135	0.61022
within recording centered force	0.30833	Not available
equal recording weight linear	0.5202	Not available

The within-recording centered slope is 0.08233 V units/N, and equal recording weighting gives 0.11616 V units/N. Both are lower than the press-weighted pooled slope. Their missing cross-validation values are intentional: the first includes recording-specific intercepts unavailable for new recordings; the second is a weighting diagnostic. Their tabulated R²/RMSE are unweighted descriptive scores for comparability. The richer adjusted model improves recording-held-out RMSE to 0.6102 V, which shows that acquisition and location information materially explain response variation. A single pooled line should therefore be interpreted as an average under this experiment rather than spatially uniform calibration.

Location, trial order, force distribution and baseline/camera settings are incompletely crossed. Their contributions cannot be uniquely identified as physical taxel effects. No taxel is removed, sign-flipped, declared the best sensor or given a separate primary force-response curve. A random-effects variance decomposition would be poorly identified with one specimen and one acquisition session; the simpler fixed-block diagnostics expose the confounding without asserting generalization to a population of skins.



Supplementary Figure S1. Spatial consistency of recording-mean residuals. Error bars are 95% t intervals across eight recording means per location.

11. Localization Performance

Learned nine-region localization

Overall model-assisted BAURods localization accuracy was **84.40%**, with **macro F1 0.8475**, across 7,154 eligible held-out contact frames from 54 recordings. This describes nine discrete regions of one sensing skin. Requiring both successful learned contact detection and the correct region gave **83.61% joint accuracy** on the same eligible positive frames. It is not a millimetre spatial-resolution result. Model report, Tables 10 and 19 [S1].

The model therefore supplies evidence of useful region discrimination on held-out recordings. Spatial/generalization differences remain material: TEST1 accuracy was 42.71%, versus 87.43-97.87% for TEST2-TEST6, and ROI 2 recall was 56.00%. No group or location was removed to improve the headline. The earlier known-contact argmax accuracy of 94.31% comes from different features, press peaks, weighting, force coverage and recording inclusion; it cannot be presented as outperforming the learned model or averaged with its score.

Archived gate and known-contact press diagnostics

Localization is evaluated separately from force-response characterization. The predicted location is the archived software's dominant ROI at the event's greatest global response, with the saved **5-V gate** retained. A below-threshold output is an explicit **No detection** result and counts as failure to localize a known press.

Overall BAURods localization accuracy was **0.597%** (95% recording-bootstrap interval 0.195-1.089%).

Macro-averaged F1 was 0.01135 (95% interval 0.00367-0.01853). Macro precision was 0.44444 and macro recall was 0.00577. There were 5,325 no-detection outcomes. Conditional accuracy among detected presses was 100.00%, but its small and selected denominator prevents use as headline localization accuracy. For comparison,

removing the optical gate and always choosing the largest ROI gave 94.31% location accuracy; this diagnostic is not the deployed rule's accuracy.

For a distinct **known-contact location-discrimination task**, ungated argmax gives macro precision 0.95221, macro recall 0.94447, and macro F1 **0.94437** over all 5,357 presses. This uses external knowledge that a press occurred and asks which ROI responded most strongly. It is an offline diagnostic, not thresholded autonomous localization. Its separate confusion matrix is in figures/spatial_diagnostics/ungated_localization_diagnostic.png. The difference between this diagnostic and end-to-end performance identifies the saved detection gate as a major limitation, without claiming that a replacement gate has been validated.

The confusion matrix includes the no-detection column and all nine target classes. Per-class precision/recall/F1 are supporting diagnostics in localization_per_class.csv, not independent sensor-characterization conclusions. The supplied technical report and saved outer predictions now support the separate frame-level learned-model results above. Those predictions have not been converted into a new press-level model score.

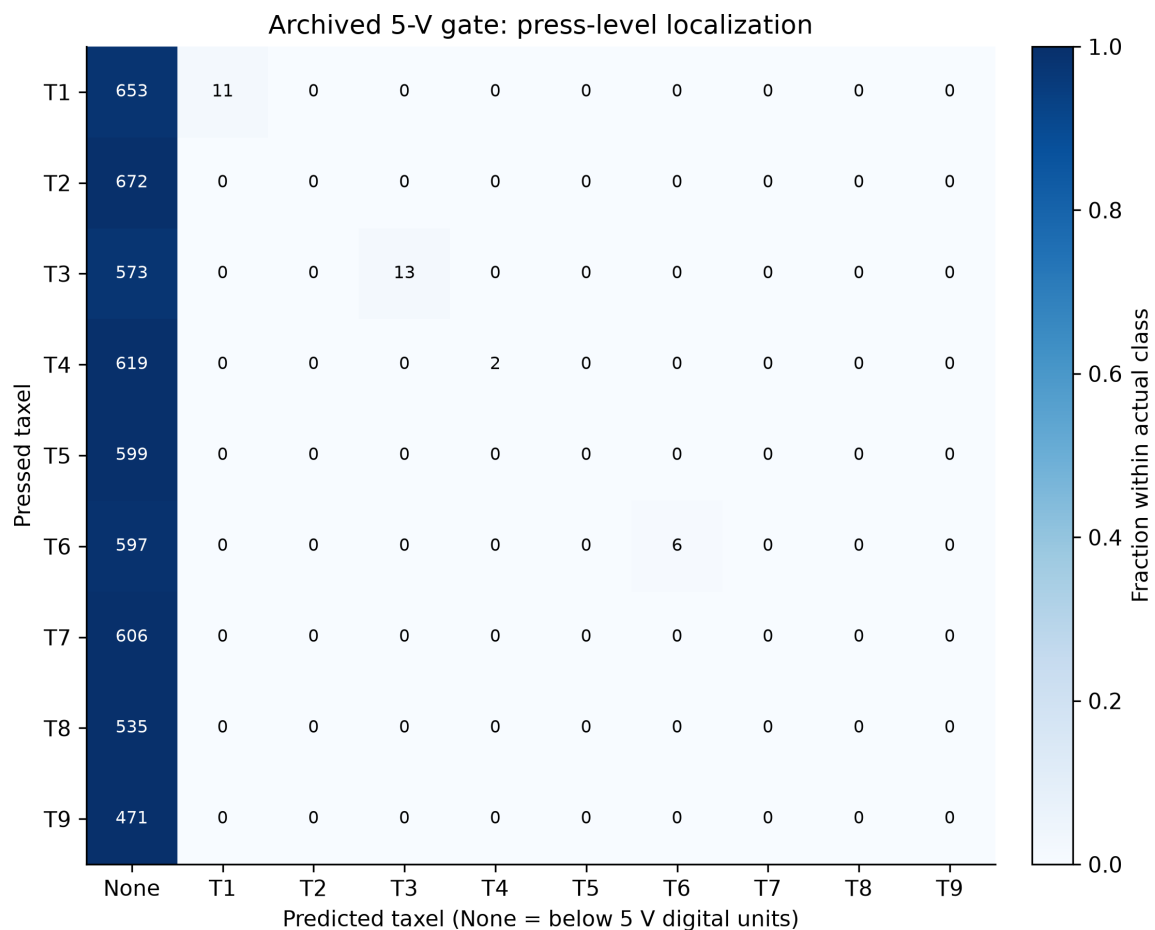


Figure 9. Archived-gate press-level localization. No-detection outcomes remain in the denominator; this matrix does not describe the later learned localization model.

12. Characterization Summary

The combined evidence supports BAURods as an integrated 3×3 visuotactile skin with measurable force-associated optical response, useful spatial information and strong retrospective model-assisted contact discrimination. The clearest application result is 98.50% contact balanced accuracy, complemented by 84.40% nine-region localization accuracy and 83.61% joint contact-and-region accuracy within the specified frame evaluation. Force estimation improved to 0.499 N MAE but remains less mature, particularly for lighter contacts.

The physical press analysis and learned prediction study answer complementary questions. The first quantifies the recorded optical behavior across the tested skin; the second measures the usefulness of richer original-video features for inference on held-out recordings. Together they support the feasibility of computational interpretation of the sensing skin under the archived manual conditions. They do not isolate which modeling component caused each gain or establish prospective live-system performance. The low recall of the earlier 5-V gate remains visible as a limitation of that acquisition configuration.

The experiment supports tested force/response ranges, global descriptive statistics, a pooled response model, apparent slope, force-conditioned dispersion, sampled baseline statistics and separately scoped detection/localization results. Intrinsic force sensitivity, a validated linear operating interval, a reliable force detection limit, intrinsic response/recovery, hysteresis, saturation capacity, long-term drift and between-device reproducibility remain **not quantitatively assessed**. Neither the 0.05-3 N model interval nor the 23.3004 N observed press maximum establishes a validated operating limit.

Reproducibility and evidence

`global_sensor_characterization.py` reads the source ZIP without modifying it. SHA-256: `e039dcd49f3390d34dc979490dcf46a6ecc43d9e6a2aec08e4952acc14b5006a`. The event table contains recording identity, force-sample row indices, start/peak/end timestamps, optical-peak row, pressed location, baseline block and all derived values. The source reconciliation table verifies exported maximum/sum arithmetic, prediction reconstruction, force-unit conversion and timing. `validation_results.json` records the executed independent arithmetic checks. `analysis_results.json`, CSV tables, PNG/PDF/SVG figures and the Excel export provide inspectable results. Source code and output hashes are recorded in `run_manifest.json`.

Method references: [SciPy peak prominence](#) defines the peak-separation statistic; [NIST calibration model validation](#) motivates residual assessment beyond R^2 ; [scikit-learn grouped cross-validation](#) explains why observations sharing a group must stay together during validation. All numerical results above come from the supplied experimental data, not those references.

The model extension is reproduced with `integrate_model_characterization.py`. It independently recomputes selected-model metrics and combined contact/region outputs from saved outer predictions, checks matched fixed/tuned frame IDs and truth, and confirms the manual archive hash. `model_report_validation.json` records the arithmetic checks and input hashes; `model_assisted_outer_predictions.csv` retains the evaluated predictions. Copied source tables and the supplied PDF are in `model_report_review/evidence/`. No models were retrained or installed into the GUI during this integration.

Appendix A. Thesis-ready Results and Discussion

The BAURods 3 × 3 visuotactile skin exhibited measurable optical responses across all nine sensing regions and supported model-assisted contact detection, region localization and force estimation. Retrospective nested grouped evaluation yielded **98.50% contact balanced accuracy**, **84.40% localization accuracy**, and **0.499 N force mean absolute error**. These outcomes establish the strongest supported computational capabilities of the integrated skin under the archived manual conditions. Global physical-response characterization complements these predictive results by describing the sensor's recorded optical behavior across 5,357 retained repeated presses. Technical report, Tables 9-11 [S1].

The model results use original, unannotated video features and six complete held-out TEST groups, with equal total weight for each recording. Force and localization use 7,154 eligible frames from 54 primary press recordings within 0.05-3 N. Contact adds 1,352 no-contact frames from 11 recordings (8,506 frames; 65 recordings total). These are correlated frame observations, not 8,506 independent contact events. The model protocol excludes 18 replay-designated recordings included in the broader 72-recording pulse analysis. Both analyses trace to the same manual archive; they are complementary evaluations, not independent replication. The recordings were collected on 8 August 2026; 9 September refers to saved model experiments, not new acquisition.

Each retained physical press contributed one force-response observation, and uncertainty was assessed by resampling complete recordings within the nine fixed locations. The 29,318 camera samples were not treated as independent physical trials. Peak force was paired with the greatest global optical response within the same force-defined interval. The primary global response was the maximum across nine ROI means of positive baseline-corrected V-channel change, followed by the maximum within each press. This response used the archived color-magnified output, whereas the learned models used multivariate original-video features. Their results consequently characterize different stages of the sensing system.

Retained press peaks covered 1.151-23.300 N, with optical responses of 0.707-8.570 V digital units. The pooled relationship was $S = 0.90855 + 0.12106 F$, corresponding to an apparent slope of 0.12106 V digital units/N (95% recording-bootstrap interval 0.08885-0.14901). The positive slope supports a force-associated optical response at the global level. Its R^2 of 0.2229 and recording-held-out R^2 of 0.1934 indicate substantial remaining dispersion. The slope is processing- and protocol-dependent and should not be treated as intrinsic material sensitivity or a transferable linear force calibration. The tested maximum does not establish sensor capacity.

The 11 no-contact recordings had a recording-equal mean response of 0.8980 V digital units and RMS temporal standard deviation of 0.0437 V units. The median press-peak excess relative to this baseline noise was 18.406. This ratio provides an exploratory measure of response contrast under the recorded processing configuration, rather than physical optical-power SNR. Global residual SD was 0.7046 V units; it describes force-conditioned dispersion under manual loading, including spatial, acquisition and loading-rate variation. Controlled repeatability and long-term drift remain unquantified.

The learned contact detector was the strongest predictive component. It attained 97.00% recall and 97.51% accuracy, with macro F1 0.9581. Balanced accuracy increased from 91.86% for the previous fixed Random Forest to 98.50% for the complete tuned selection procedure. No false positives were observed on the 11 dedicated no-contact recordings. This supports effective discrimination of the evaluated contact and unloaded conditions, while the small number and limited diversity of controls restrict generalization to sustained live operation. These are recording-balanced frame scores and do not establish event-onset detection or a force detection limit. Technical report, Tables 11 and 14 [S1].

Nine-region localization accuracy improved from 73.79% to 84.40%, with macro F1 increasing from 0.7400 to 0.8475. Requiring both a positive contact decision and the correct region yielded 83.61% joint accuracy on eligible positive frames. These findings support useful spatial discrimination by the complete sensing skin without treating the regions as separate sensors. Performance varied across held-out groups and locations: TEST1 reached 42.71% accuracy and ROI 2 recall was 56.00%. The overall gain therefore coexists with spatial and acquisition-dependent

weaknesses. The regions are discrete labels; no continuous localization error in millimetres was measured. Technical report, Tables 10, 15-16 and 19 [S1].

Force estimation benefited from the expanded modeling procedure. MAE decreased from 0.637 to 0.499 N, a 21.68% reduction, while RMSE decreased from 0.770 to 0.632 N. R^2 increased from -0.030 to 0.306, and bias changed from -0.289 to +0.078 N. This supports improved retrospective force inference within 0.05-3 N, although force estimation remains the least mature quantitative function. Error increased in all four diagnostic bins below 2 N, reaching 1.241 N MAE below 0.5 N. The 0.05 N inclusion boundary is not a validated detection limit, and 3 N is not a maximum operating specification. Technical report, Tables 9 and 17 [S1].

The archived acquisition gate was evaluated separately to explain the behavior of the earlier software configuration. Its fixed 5-V digital-unit criterion detected only 32 of 5,357 press events (0.597% recall); all 32 were correctly localized. When contact was externally known and the gate was removed, the strongest-ROI rule achieved 94.31% press-level accuracy and macro F1 0.94437. This indicates that spatial information was present even when the configured gate suppressed outputs. The later learned detector provides a separate demonstration of contact discrimination from richer features. Because input processing, eligibility, weighting and evaluation units differ, these numbers are not a direct before/after comparison of the gate and learned detector.

Offline stateful model inference took a median 41.10 ms per frame (95th percentile 44.46 ms), covering the model tasks and tabular preprocessing but excluding acquisition, decoding, pixel extraction and display. This is a computational timing measurement, not sensor response time. Intrinsic response/recovery and hysteresis were not quantitatively assessed: manual presses lacked controlled loading/release conditions, optical sampling was sparse, and the archived primary response underwent temporal filtering. The saved causal model also uses optical history and smoothing, whose onset/release effects require separate measurement. Technical report, Tables 21 and 23 [S1].

Overall, the combined results support BAURods as a sensing skin with useful optical and spatial information that can be interpreted computationally for contact detection, region classification and exploratory force estimation. Contact discrimination and localization provide the strongest demonstrated capabilities. Further work should prioritize light-force estimation, acquisition robustness and prospective validation with frozen models. The reported evidence remains retrospective, comes from one skin and collection session, and does not establish superiority over other sensors or validated live deployment.

Appendix B. Source register and validation

[S1] Manual Calibration Models Technical Report for Thesis Writing, prepared 5 October 2026. Tables 9-11 document the principal nested model results; Table 19 the combined pipeline; Table 23 offline inference timing. This report is supplied by the researcher and traces to the same manual archive.

[S2] Manual Calibration (2).zip. SHA-256:

e039dcd49f3390d34dc979490dcf46a6ecc43d9e6a2aec08e4952acc14b5006a. The primary physical analysis uses the archived synchronized measurements and raw reference force.

[S3] Saved fixed and tuned model runs: retraining_20260909 and tuning_20260909. The model extension independently recomputes selected outer-prediction metrics, checks matching held-out frame identities and truth, and retains the original evaluation weights.

[S4] Global analysis outputs: global_force_response_data.csv, fitted_model_comparison.csv, baseline_statistics.csv, contact_detection_metrics.csv, localization_per_class.csv, and the spatial diagnostics. Complete numerical evidence and reproducible scripts accompany the analysis package.

V means the OpenCV HSV brightness channel in image digital units, not volts. N denotes newtons; MAE, mean absolute error; RMSE, root mean squared error; ROI, region of interest; FPR, false-positive rate; CI, confidence interval; SD, standard deviation.

Verification: all 5,357 retained event maxima were checked against source measurements. The model extension reproduced 36 principal and combined metrics, with a maximum absolute difference of 2.34×10^{-15} . The 17-sheet numerical workbook was checked across 212,817 cells with zero mismatches. These checks establish numerical consistency, not independent physical or prospective validation.

Figure numbering follows the reproducible analysis package. Figure 10 appears first to introduce the model-assisted findings; Figures 1-9 retain their original physical-characterization numbering. Wide numerical tables are split into labelled parts to preserve legibility and all reported values.